



**University of
Nottingham**

UK | CHINA | MALAYSIA

Faculty of Science

**Year 3 Work Based Project
Project Agreement**

Academic Year 2025-26

Apprentice: Yahya Hafeji

Employer: Northamptonshire Healthcare NHS Foundation Trust

Manager: Edward Buxton

In submitting this document, you confirm that you have discussed this project with your manager and have their support for the project, and the resource and data that you require in order to complete it.

Contents

Organisational Need	2
Project Overview and Scope	2
Data Requirements	3
Tools and Computing Resources.....	3
Knowledge Skills and Behaviours.....	3

Organisational Need

Northamptonshire Healthcare NHS Foundation Trust (NHFT) delivers over 500 services across community, mental health, and specialist care pathways. Each service experiences fluctuating demand patterns, seasonal pressures, and variable staffing capacity, but currently there is no unified, data-driven framework for forecasting demand or planning capacity. Operational managers rely heavily on retrospective reports or manual spreadsheets, which limits proactive decision-making and the ability to optimise resources across services.

The organisation therefore needs a scalable Demand and Capacity Modelling Platform that can:

- Provide forecasted service demand based on historical trends and seasonal patterns.
- Model capacity scenarios to identify bottlenecks, breach risks, and staffing shortfalls.
- Enable evidence-based workforce and clinic planning to improve utilisation, reduce breaches, and enhance patient access.
- Offer a standardised, repeatable approach that can be configured for multiple service types, reducing duplication of analytical effort across the Trust.

By investing in this project, NHFT will gain a return on investment through better operational planning, reduced reliance on ad-hoc analyst time, and potential savings from optimised staffing and reduced agency costs. The solution will also support strategic planning across Divisions by improving transparency and consistency in performance management.

Project Overview and Scope

This project will design, develop, and implement a generic, configurable Demand and Capacity Modelling Platform that can be applied across NHFT's services.

In Scope

- Development of a Python-based modelling framework with modular components for:
 - Data ingestion, validation, and transformation.
 - Statistical and machine learning-based demand forecasting (e.g., ARIMA, ETS, Gradient Boosting).
 - Capacity modelling using service templates, staff rosters, and resource constraints.
 - Queueing and simulation models to estimate waiting times and service utilisation.
 - A Dash-based visual interface for running forecasts, scenarios, and simulations interactively.
- Creation of configurable service profiles, allowing each service to run tailored models without code modification.
- Integration with existing NHFT BI data sources (referrals, appointments, rosters).
- Deployment of a prototype covering a sample of representative services (e.g., Adult Dietetics, Community Nursing, and Mental Health Outpatients).

Out of Scope

- Power BI consideration.
- Full-scale deployment across all services (beyond pilot phase).
- Predictive workforce rostering or AI optimisation beyond baseline capacity modelling.
- Mobile or external-facing application access.

Data Requirements

The project will use routinely collected NHFT datasets available through the Business Intelligence (BI) environment:

- **Referrals data:** referral counts, dates, priority levels, and source of referral.
- **Appointments data:** scheduled appointments, outcomes, cancellations, DNA rates, and duration.
- **Workforce data:** staff rosters, WTEs, job roles, and session availability.
- **Service metadata:** service names, site locations, archetype classification (inpatient/outpatient/community).
- **Calendar data:** public holidays, term times, and seasonal markers for regression features.

All data will be pseudonymised and stored within NHFT's secure environment. A Data Protection Impact Assessment (DPIA) will be completed as part of the Ethics and Legal deliverable.

Tools and Computing Resources

Resource Type	Details
Programming Environment	Python (Maybe R)
Libraries	Pandas, NumPy, Scikit-learn, Statsmodels, Plotly/Dash
Database Access	NHFT SQL Server - ISEVSQLMIS-REP
Version Control	GitHub
Development Environment	Visual Studio Code (locally) or NHFT-approved virtual desktop
Compute Resources	Local execution for prototypes; potential use of NHFT Virtual Machines or Power BI Service for visualisation
Documentation & Logs	OneNote for logbook and SharePoint for documentation

Knowledge Skills and Behaviours

Code	Description	Application in Project
S1	Identify and clarify organisational problems, reformulate them into data science questions, apply scientific methods, and seek stakeholder feedback.	Translating operational planning challenges into data science models; consulting with service managers to define requirements and validate outputs.
S2	Perform data engineering: create, transform, and store datasets with governance and quality control.	Developing reproducible pipelines for referral and appointment data, applying validation and data quality checks.
S3	Use programming languages and tools for analysis, visualisation, and integration, adhering to software development standards.	Building the modular platform in Python with version control, documentation, and reusable functions.
S4	Build models, validate results with statistical testing, and use appropriate ML/optimisation techniques.	Implementing forecasting, optimisation, and simulation components to model demand and capacity.

S5	Implement data solutions and evaluate architectures, scalability, and value for money.	Designing a framework that can scale to multiple services with minimal configuration and maintenance.
S6	Communicate and disseminate outputs effectively using dashboards and storytelling.	Delivering a Dash application and presentation to demonstrate organisational insights.
S8	Use project delivery techniques and manage resources effectively.	Applying Kanban methodology with milestones aligned to project deliverables.
B4	Consider problems in organisational context.	Aligning the model to NHFT's operational planning and performance improvement goals.
B5	Maintain scientific integrity and impartiality.	Applying rigorous statistical testing and transparent reporting of assumptions and limitations.